Laboratorio 3: Algoritmos de enrutamiento

Parte 2. Evidencias de la implementación en Packet Tracer

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1. Tablas de enrutamiento

Salidas de `show ip route` en los cinco routers, tomadas con OSPF (área 0) y EIGRP (AS 100) en operación, y con la redistribución mutua y la sumarización configuradas en R-CENTRAL. Se omite la leyenda de códigos al inicio de cada salida.

1.1 R-OSPF-1

Hacia 192.168.0.0/24 hay dos rutas de igual costo. Todo el dominio EIGRP llega como una sola ruta externa: O E1 172.16.0.0/24.

```
R-OSPF-1#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/100] via 10.0.0.6, 00:00:18, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/70] via 10.0.0.6, 00:00:18, Serial0/0/1
O    192.168.0.0/24 [110/101] via 10.0.0.2, 00:00:18, Serial0/0/0
                    [110/101] via 10.0.0.6, 00:00:18, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

1.2 R-OSPF-2

Dos rutas de igual costo hacia cada VLAN de R-OSPF-1 y la misma ruta resumen O E1 172.16.0.0/24.

```
R-OSPF-2#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/100] via 10.0.0.10, 00:03:36, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/70] via 10.0.0.10, 00:03:36, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/101] via 10.0.0.1, 00:03:26, Serial0/0/0
                       [110/101] via 10.0.0.10, 00:03:26, Serial0/0/1
O       192.168.1.64/28 [110/101] via 10.0.0.1, 00:03:26, Serial0/0/0
                        [110/101] via 10.0.0.10, 00:03:26, Serial0/0/1
```

1.3 R-CENTRAL

Conoce las redes de ambos dominios. Las rutas hacia Null0 son los resúmenes que anuncia: 172.16.0.0/24 hacia OSPF y 192.168.0.0/23 hacia EIGRP.

```
R-CENTRAL#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/150] via 10.0.0.9, 00:04:36, Serial0/0/1
                    [110/150] via 10.0.0.5, 00:04:36, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/307200] via 10.0.0.17, 00:04:51, Serial0/1/0
                     [90/307200] via 10.0.0.21, 00:04:47, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/153856] via 10.0.0.21, 00:04:47, Serial0/1/1
D       172.16.0.128/26 [90/156160] via 10.0.0.17, 00:04:51, Serial0/1/0
D       172.16.0.192/27 [90/156160] via 10.0.0.17, 00:04:51, Serial0/1/0
D       172.16.0.224/28 [90/156160] via 10.0.0.17, 00:04:51, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:04:36, Null0
O    192.168.0.0/24 [110/51] via 10.0.0.9, 00:04:36, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/51] via 10.0.0.5, 00:04:36, Serial0/0/0
O       192.168.1.64/28 [110/51] via 10.0.0.5, 00:04:36, Serial0/0/0
```

1.4 R-EIGRP-1

El dominio OSPF llega como una sola ruta, 192.168.0.0/23. Las redes de enlace del lado OSPF entran como externas (D EX).

```
R-EIGRP-1#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/977920] via 10.0.0.18, 00:05:55, Serial0/0/1
D EX    10.0.0.4/30 [170/977920] via 10.0.0.18, 00:06:06, Serial0/0/1
D EX    10.0.0.8/30 [170/977920] via 10.0.0.18, 00:06:09, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/849920] via 10.0.0.18, 00:06:06, Serial0/0/1
                     [90/977920] via 10.0.0.14, 00:06:06, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 7 subnets, 5 masks
D       172.16.0.0/25 [90/281856] via 10.0.0.14, 00:06:06, Serial0/0/0
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
C       172.16.0.224/28 is directly connected, GigabitEthernet0/0.30
L       172.16.0.225/32 is directly connected, GigabitEthernet0/0.30
D    192.168.0.0/23 [90/2866176] via 10.0.0.18, 00:05:55, Serial0/0/1
```

1.5 R-EIGRP-2

Hacia cada VLAN de R-EIGRP-1 hay dos rutas con métricas distintas (balanceo desigual).

```
R-EIGRP-2#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/977920] via 10.0.0.22, 00:06:48, Serial0/0/1
D EX    10.0.0.4/30 [170/977920] via 10.0.0.22, 00:06:59, Serial0/0/1
D EX    10.0.0.8/30 [170/977920] via 10.0.0.22, 00:06:59, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/849920] via 10.0.0.22, 00:06:59, Serial0/0/1
                     [90/977920] via 10.0.0.13, 00:06:59, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/284160] via 10.0.0.13, 00:06:59, Serial0/0/0
                        [90/852480] via 10.0.0.22, 00:06:59, Serial0/0/1
D       172.16.0.192/27 [90/284160] via 10.0.0.13, 00:06:59, Serial0/0/0
                        [90/852480] via 10.0.0.22, 00:06:59, Serial0/0/1
D       172.16.0.224/28 [90/284160] via 10.0.0.13, 00:06:59, Serial0/0/0
                        [90/852480] via 10.0.0.22, 00:06:59, Serial0/0/1
D    192.168.0.0/23 [90/2866176] via 10.0.0.22, 00:06:48, Serial0/0/1
```

2. Balanceo de carga en OSPF (costo igual)

Entre R-OSPF-1 y R-OSPF-2 existen dos caminos. El costo de las interfaces seriales se fijó con `ip ospf cost` para que ambos sumen lo mismo:

Camino de R-OSPF-1 hacia 192.168.0.0/24	Suma de costos	Total
Enlace directo	Se0/0/0 (100) + LAN de R-OSPF-2 (1)	101
Por R-CENTRAL	Se0/0/1 (50) + Se0/0/1 de R-CENTRAL (50) + LAN de R-OSPF-2 (1)	101

Con el mismo costo, OSPF instala ambas rutas y reparte el tráfico entre ellas. En la salida de `show ip ospf interface` de R-OSPF-1 se observan los costos 100 y 50 en las seriales, la adyacencia con 2.2.2.2 y 3.3.3.3, y las subinterfaces de las VLAN como pasivas.

```
R-OSPF-1#show ip ospf interface
GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.65
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
```

```
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 100
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 50
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

Rutas resultantes en la tabla de R-OSPF-1:

```
R-OSPF-1#show ip route
...
O    192.168.0.0/24 [110/101] via 10.0.0.2, 00:00:18, Serial0/0/0
                    [110/101] via 10.0.0.6, 00:00:18, Serial0/0/1
...
```

3. Balanceo de carga en EIGRP (75/25)

Se ajustaron `bandwidth` y `delay` para que el camino por R-CENTRAL tenga el triple de métrica que el enlace directo, y se habilitó `variance 3` en R-EIGRP-1 y R-EIGRP-2. Rutas de R-EIGRP-2 hacia 172.16.0.128/26 (VLAN 10 de R-EIGRP-1):

Ruta	Siguiente salto	Métrica	Distancia reportada	Tráfico
Successor (enlace directo)	10.0.0.13, Se0/0/0	284,160	28,160	75 %
Feasible successor (por R-CENTRAL)	10.0.0.22, Se0/0/1	852,480	156,160	25 %

La ruta por R-CENTRAL cumple la condición de factibilidad porque su distancia reportada (156,160) es menor que la métrica del successor (284,160). Como 852,480 = 3 × 284,160, queda dentro de `variance 3` y ambas rutas se instalan en la tabla; el tráfico se reparte en proporción 3:1, es decir, 75 % y 25 %. Lo mismo ocurre con 172.16.0.192/27 y 172.16.0.224/28.

Salida de `show ip eigrp topology` en R-EIGRP-2 (extracto):

```
R-EIGRP-2#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(5.5.5.5)

P 172.16.0.128/26, 2 successors, FD is 284160
         via 10.0.0.13 (284160/28160), Serial0/0/0
         via 10.0.0.22 (852480/156160), Serial0/0/1
P 172.16.0.192/27, 2 successors, FD is 284160
         via 10.0.0.13 (284160/28160), Serial0/0/0
         via 10.0.0.22 (852480/156160), Serial0/0/1
P 172.16.0.224/28, 2 successors, FD is 284160
         via 10.0.0.13 (284160/28160), Serial0/0/0
         via 10.0.0.22 (852480/156160), Serial0/0/1
...
```

4. Conectividad de extremo a extremo

Ping y tracert desde PC-OSPF1-V10 (192.168.1.10, VLAN 10 de R-OSPF-1) hacia PC-EIGRP2-LAN (172.16.0.10, LAN de R-EIGRP-2). El ping recibe 4 de 4 respuestas, sin pérdida de paquetes. El tracert pasa por R-OSPF-1 (192.168.1.1), R-CENTRAL (10.0.0.6) y R-EIGRP-2 (10.0.0.21) hasta llegar al host destino.

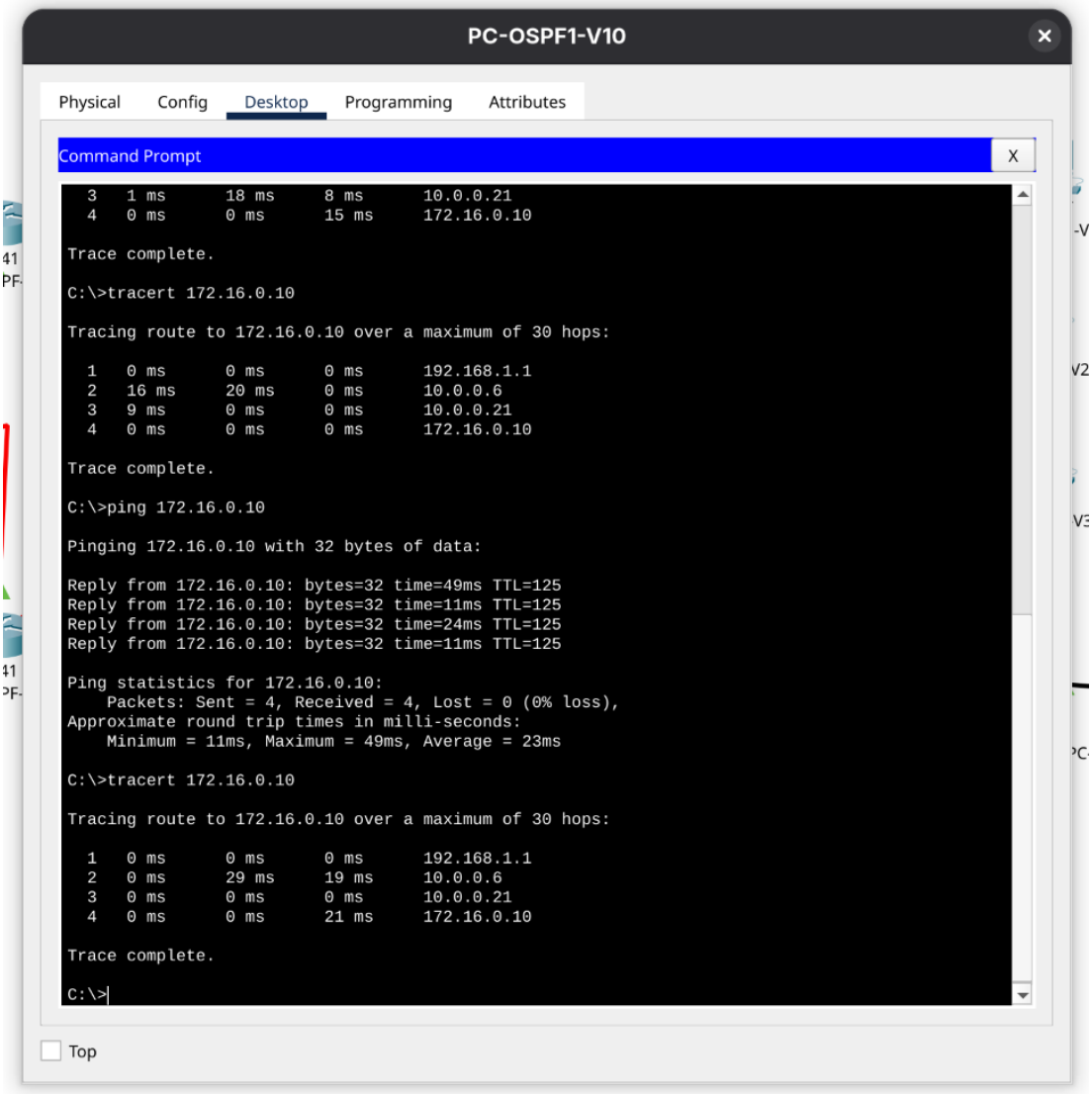


Figura 1. Ping y tracert desde PC-OSPF1-V10 hacia PC-EIGRP2-LAN.